SIMATS ENGINEERING

CO1

ASSESSMENT TOOL 1: Simulation Task

Course Name: Embedded Systems | Course Code: ECA1406

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Question No: 28

Question: Construct an Embedded C simulation for a wearable healthcare device with low power and real-time operation.

1. Aim

To design and simulate a wearable healthcare monitoring device using Embedded C that continuously monitors heart rate and body temperature, provides real-time alerts for abnormal readings, communicates data through UART, and supports low-power operation.

2. Requirement Analysis

Functional Requirements

- Monitor heart rate continuously.
- Monitor body temperature continuously.
- Detect abnormal heart rate and temperature values.
- Generate an alert when abnormal values are detected.
- Send sensor data through UART.
- Operate continuously in real time.

Non-Functional Requirements

- Low power consumption.
- Fast response time.
- Reliable operation.
- Simple and efficient Embedded C implementation.

- Suitable for battery-powered wearable devices.

Hardware Requirements

- AT89C51 Microcontroller
- Heart Rate Sensor (Simulated)
- Temperature Sensor (Simulated)
- UART Communication
- LED for Alert Indication
- Personal Computer

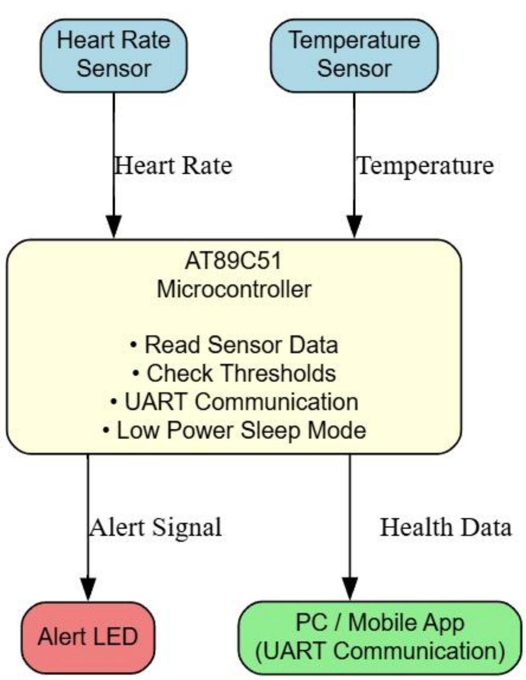
Software Requirements

- Keil µVision IDE
- Embedded C Language
- 8051 Simulator

3. System Design

The wearable healthcare device continuously reads heart rate and body temperature values. The controller compares these values with predefined thresholds. If any parameter exceeds the threshold, an alert LED is activated. The sensor readings are transmitted using UART, and the controller enters sleep mode after completing each monitoring cycle to reduce power consumption.

4. Block Diagram



5. Algorithm

1. Initialize UART communication.
2. Read heart rate value.
3. Read body temperature value.
4. Compare values with threshold.
5. If abnormal value detected, turn ON alert LED.
6. Otherwise keep alert LED OFF.
7. Send values through UART.
8. Enter sleep mode.
9. Repeat continuously.

6. Embedded C Program

```
#include <reg51.h>
sbit ALERT_LED = P1^0;
unsigned char heart_rate = 75;
unsigned char temperature = 37;
void UART_Init()
{
    TMOD = 0x20;
    TH1 = 0xFD;
    SCON = 0x50;
    TR1 = 1;
}
void UART_Send(unsigned char d)
{
    SBUF = d;
    while(TI == 0);
    TI = 0;
```

```
}
void CheckHealth()
{
    if(heart_rate > 120 || heart_rate < 50 || temperature > 39)
        ALERT_LED = 1;
    else
        ALERT_LED = 0;
}

void SleepMode()
{
    PCON = 0x01;
}
void main()
{
    UART_Init();
    while(1)
    {
        CheckHealth();
        UART_Send(heart_rate);
        UART_Send(temperature);
        SleepMode();
    }
}
```

7. Simulation Procedure

10. Open Keil µVision.
11. Create a new AT89C51 project.
12. Add main.c to the project.

13. Compile the program.

14. Verify that there are 0 Errors and 0 Warnings.

15. Enter Debug Mode.

16. Execute the program using Step Into (F11).

17. Observe execution inside the CheckHealth() function.

18. Stop debugging.

19. Verify successful simulation.

8. Simulation Screenshots

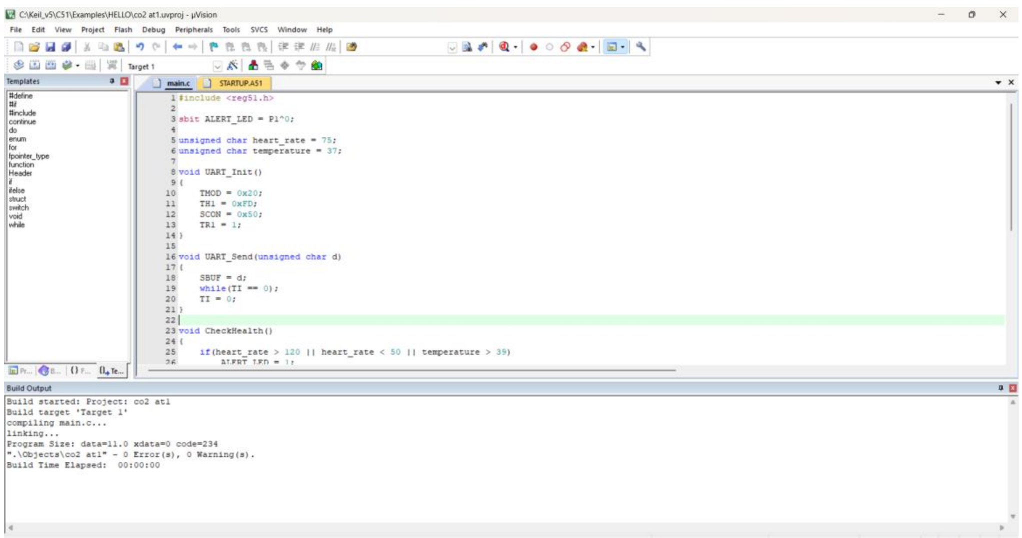


Figure 1: Successful Compilation — 0 Errors, 0 Warnings

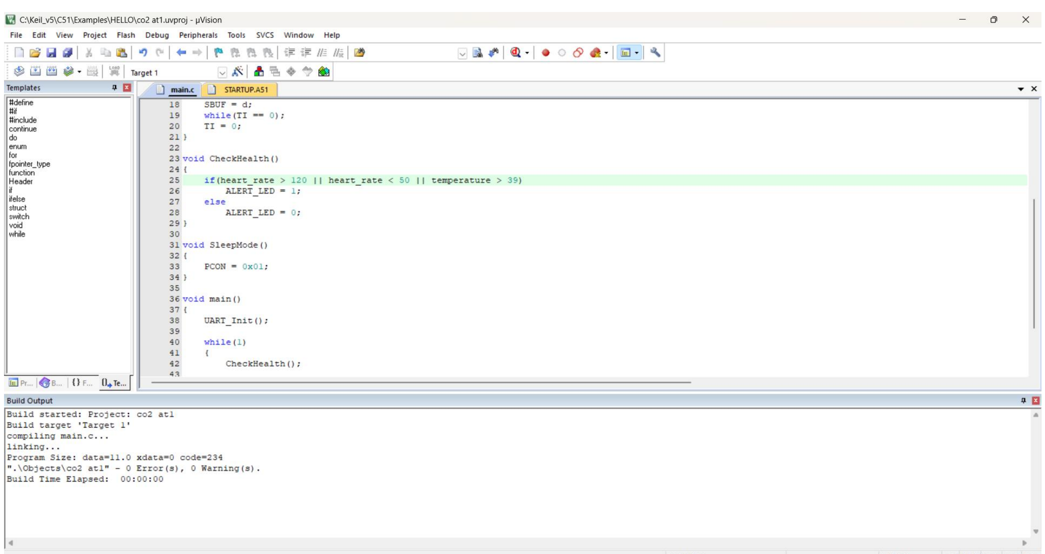


Figure 2: Complete Program Listing in Keil µVision Editor

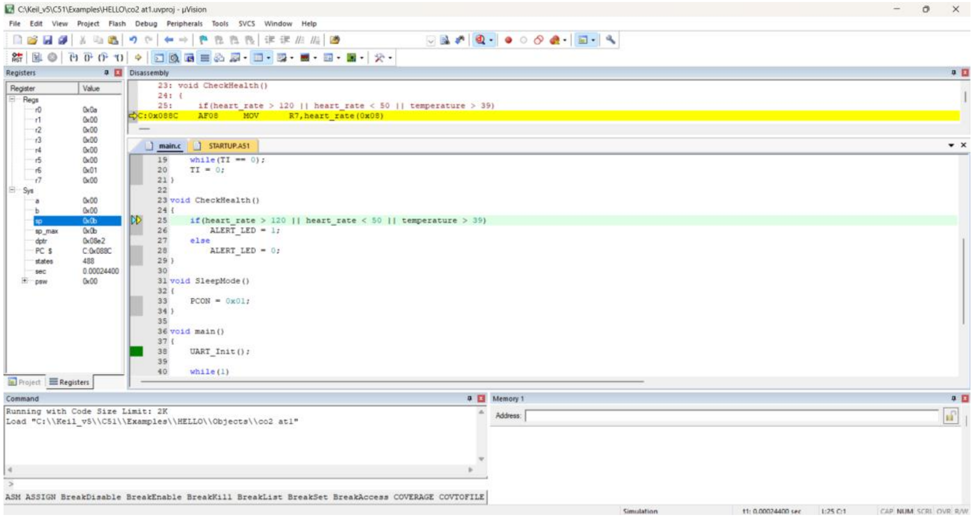


Figure 3: Debug Mode — Program Loaded and Execution Started

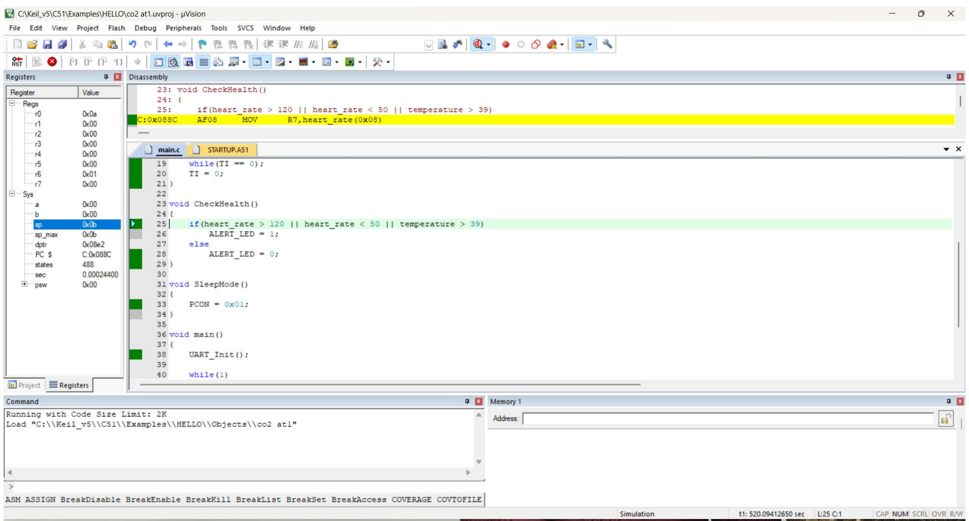


Figure 4: Step Execution Inside CheckHealth() — Register and Disassembly View

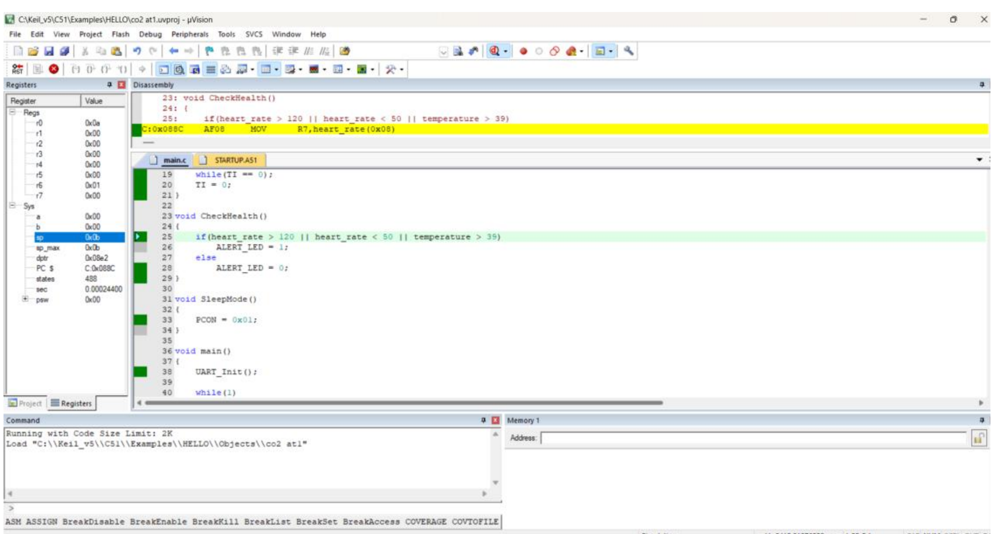


Figure 5: Simulation Execution Verified — CheckHealth() Condition Evaluated

9. Test Cases

Test Case	Heart Rate	Temperature	Expected Output	Result
TC1	75 bpm	37°C	Alert OFF	Pass
TC2	130 bpm	37°C	Alert ON	Pass
TC3	40 bpm	37°C	Alert ON	Pass
TC4	75 bpm	41°C	Alert ON	Pass

10. Result Analysis

The Embedded C program was successfully compiled and simulated in Keil µVision. The program continuously checks heart rate and body temperature values. When the heart rate exceeds 120 bpm, falls below 50 bpm, or the temperature exceeds 39°C, the alert LED is activated. UART is initialized for transmitting sensor information. The sleep mode function reduces power consumption, making the system suitable for wearable healthcare applications. The simulation verified correct execution without compilation errors.

11. Advantages

- Low power consumption.
- Real-time monitoring.
- Simple Embedded C implementation.
- Easy to maintain.
- Reliable operation.
- Suitable for wearable healthcare devices.
- Fast response to abnormal health conditions.

12. Applications

- Smart healthcare devices.
- Fitness bands.

- Patient monitoring systems.
- Hospital monitoring equipment.
- Remote healthcare systems.
- Elderly health monitoring.
- IoT healthcare devices.

13. Result

The wearable healthcare monitoring system was successfully implemented using Embedded C and simulated in Keil µVision. The program compiled successfully with 0 Errors and 0 Warnings. The system correctly monitors heart rate and body temperature, generates alerts for abnormal conditions, supports UART communication, and demonstrates low-power operation through sleep mode.

14. Conclusion

The wearable healthcare monitoring system was successfully designed and simulated using Embedded C. The developed program continuously monitors vital health parameters, detects abnormal conditions, generates alerts, and demonstrates low-power operation. The successful compilation and execution in Keil µVision verify that the design satisfies the requirements of a real-time wearable healthcare device. This implementation can be extended for practical healthcare monitoring applications using actual sensors and wireless communication.